

The Global Tea and Organic Juice companies have merged. The following information has been collected for the “Consolidation Project.”

Activity	Description	Predecessor	<i>a</i> opt	<i>m</i> ml	<i>b</i> pess
1	Codify accounts	None	16	19	28
2	File articles of unification	None	30	30	30
3	Unify price and credit policy	None	60	72	90
4	Unify personnel policies	None	18	27	30
5	Unify data processing	1	17	29	47
6	Train accounting staff	1	4	7	10
7	Pilot run data processing	5	12	15	18
8	Calculate P & L and balance sheet	6, 7	6	12	24
9	Transfer real property	2	18	27	30
10	Train salesforce	3	20	35	50
11	Negotiate with unions	4	40	55	100
12	Determine capital needs	8	11	20	29
13	Explain personnel policies	11	14	23	26
14	Secure line of credit	9, 12	13	16	19
15	End	10, 13, 14	0	0	0

1. Compute the expected time for each activity.
2. Compute the variance for each activity.
3. Compute the expected project duration.
4. What is the probability of completing the project by day 112? Within 116 days?
5. What is the probability of completing “Negotiate with unions” by day 90?

The expected times and variances for the project activities follow. What is the probability of completing the project in 25 periods?

ID	Description	Predecessor	t_e	Variance $[(b - a)/6]^2$
1	Pilot production	None	6	3
2	Select channels of distrib.	None	7	4
3	Develop mktg. program	None	4	2
4	Test market	1	4	2
5	Patent	1	10	5
6	Full production	4	16	10
7	Ad promotion	3	3	2
8	Release	2, 5, 6, 7	2	1

Activity	Identity	Predecessors	Estimated time		
			a	m	b
Investigate customer demands	A	-	1	3	4
Develop pricing strategy	B	-	1	1	2
Design product	C	-	4	5	9
Analyze project cost	D	A	1	1	1
Build prototype	E	C	4	6	12
Analyze product cost	F	E	1	1	2
Analyze product price	G	B,D,F	1	2	3
Field test	H	G	6	8	10

a) Draw AON and AOA network for the project.

b) Identity critical path(s) and critical activities.

c) Identify the probability to complete the project if:

- Customer requests that project must complete within 22 weeks.
- Customer expects that project will complete within defined expected project time.
- Project takes more 28 weeks to complete.

d) Identify the scheduled project duration if the probability to complete project is:

- 30%
- 70%